UPLB IDSC-FANS KNOWLEDGE HUB: A KNOWLEDGE MANAGEMENT SYSTEM FOR FOOD AND NUTRITION SECURITY

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Introduction

The UPLB IdSC-FaNS Knowledge Hub is a web-based Knowledge Management System built specifically for the UPLB Interdisciplinary Studies Center on Food and Nutrition Security. It serves as a centralized repository for all research under food and nutrition security and allows researchers and policymakers to engage in community discussions. AI research insights and analytical tools in the web app help in converting explicit knowledge from the research available to tacit knowledge in order to gain more meaningful insights on the current status of food and nutrition research in the Philippines.

Methodology

Tech Stack:

Frontend: Next.js, React, Tailwind CSS
Backend: Node.js, Express, Socket.io
Database: MongoDB, Elasticsearch
AI: Google Gemini 2.5 Flash

Evaluation:

- N = 26 participants across three role-branched tracks (20 Standard Users, 5 Organization Admins, 1 Site Admin) completed a fully remote, unmoderated, mixed-methods evaluation via Google Forms
- - 23 tasks assessed using seven instruments: SUS, task-success rate, SEQ, feature-specific Likert grids, AI insight accuracy, AI extraction accuracy, and open-ended thematic feedback

Results

83.17
SUS
Good - Exellent band

22/23
at
≥ 95% and ≤ 1.5 SEQ

86–100%
AI metadata extraction accuracy per field

- AI metadata extraction most useful feature
- Feature-satisfaction Likerts: 4.69 – 4.96 / 5
- System endorsement ("valuable for UPLB research community"): 4.92 / 5, lowest spread (SD = 0.27)

Conclusion

- All five research objectives were met at both the satisfaction and task-success layers
- AI metadata extraction was the most-named useful feature in the open-ended responses
- Participants endorsed the system as valuable for the UPLB research community, with several saying they wished it had existed during their own studies